EXECUTIVE SUMMARY

VendorPulse AI

A GenAI copilot for Vendor DSD scheduling, vendor scoring and arrival-delay / reschedule prediction. Team: TEAM012 · Member: V V S S N Padmaji Saladi (2025mb26046) · Instructor: Vinaya Sathyanarayana

Problem

A Mid-West supercenter chain (about 300 stores) receives many products through Vendor Direct Store Delivery (DSD). Large vendors send an Advanced Shipping Notice (ASN), but small and local vendors follow the NEX process: no ASN, irregular arrival times, and reschedules or no-shows with little notice. Receiving leads plan dock staffing from memory, phone calls and spreadsheets.

- Unplanned arrivals find the dock unstaffed, so vendors scan their own pallets. This causes miscounts, variances and shrink.
- Associates wait for vendors who arrive late or do not come, which wastes receiving labor.
- Vendor reliability is judged informally, so chronically late vendors are not raised with Merchandising.
- Years of delivery history exist but are not used to plan the day.

Target user

Primary: store receiving leads and store directors who plan dock labor each morning. **Secondary:** receiving associates on Zebra handhelds at the dock, and Merchandising / Vendor Management teams who review vendor reliability.

Product concept

VendorPulse AI is a receiving copilot for small and local DSD vendors. For one store it shows which vendors are expected today and this week, a predicted arrival window, the risk that each vendor is late or reschedules, and a 0 to 100 reliability score. Users can ask questions in plain English on a handheld ("Which vendors are likely to be late this afternoon?"). Every answer shows its evidence and a confidence level. The copilot also reads vendor emails and dock notes and proposes schedule changes that a person confirms.

Why AI

A dashboard can list past deliveries; it cannot estimate what will happen this afternoon. Agreed delivery windows (the rule-based option) do not hold for NEX vendors. In our simulated test, the agreed window placed only 49% of arrivals within 45 minutes; a simple vendor average reached 63% and the ML model 70%. The risk bands carry real signal: 55% of High-risk deliveries were actually late or rescheduled versus 11% of Low-risk ones. An LLM adds two things a dashboard cannot: answering free-form questions on a handheld, and reading unstructured vendor messages. The LLM never calculates a prediction or score. It calls the ML models and data services and explains their output.

Recommendation: Pilot

Run a **10-week pilot in one store** with real NEX delivery data, then decide on scale. Do not scale yet:

- The prototype proves the workflow, the AI role and the trust model, but its accuracy figures come from simulated data.

- Arrival accuracy (70%) is below the 80% target set at topic approval, so real data and Phase 2 signals (weather, traffic) must close that gap.
- Scale only if the pilot meets the go/no-go thresholds: 80% arrival accuracy, at least 90% grounded answers with zero invented numbers, fewer unstaffed arrivals and self-scan events, and store users who trust and use it.

Approach: API-first LLM (Azure OpenAI, aligned with the retailer's Microsoft stack) plus in-house ML models on Azure ML, because the value is in the retailer's own delivery history, not in a custom language model. Humans keep every staffing, schedule-change and vendor decision.